

Dr. Doc:

Exam: Gross Anatomy

Year: 2014 E.C

Dr. Doc Tutorial

Which muscles are normally active during quiet inspiration?

- A. External and internal intercostal muscles
- B. External intercostal muscles and diaphragm
- C. Internal intercostal muscles and diaphragm
- D. External and internal intercostal muscles and diaphragm

The correct answer is B. External intercostal muscles and diaphragm.

Quiet (normal, resting) inspiration is an active process driven by two main groups of muscles:

The Diaphragm (Primary Muscle):

- * Contraction of the diaphragm pulls its dome downward toward the abdominal cavity.
- * This increases the vertical dimension of the thoracic cavity and accounts for approximately 75% of the air movement during quiet breathing at rest.

The External Intercostal Muscles:

- * Contraction of these muscles elevates the ribs upward and outward (the "bucket-handle" and "pump-handle" movements).

* This increases the anteroposterior and lateral dimensions of the chest cavity.

Why the other options are incorrect:

- Internal Intercostal Muscles are muscles of expiration (specifically during forced or active expiration, such as during exercise or coughing), where they pull the ribs downward and inward to compress the thoracic cavity.

- Therefore, any option containing the internal intercostal muscles (A, C, and D) is incorrect for quiet inspiration.

The lungs are attached to the walls of the thorax by means of the:

A. Pericardial membrane

B. Parietal pleural membrane

C. Visceral pleural membrane

D. Mesentery

E. Diaphragm

The correct answer is B. Parietal pleural membrane.

To answer this question correctly, you must distinguish between the two layers of the pleural cavity:

1. Parietal Pleural Membrane (Option B)

- *Anatomy: The parietal pleura is the outer, thick serous membrane that lines the internal surface of the thoracic wall (chest wall), the upper surface of the diaphragm, and the lateral sides of the mediastinum.*

- *Function: It is physically attached to the thoracic wall via a thin layer of connective tissue called the endothoracic fascia.*

2. Visceral Pleural Membrane (Option C)

- *Anatomy: The visceral pleura is the inner serous membrane that directly covers and adheres tightly to the outer surface of the lungs, dipping into its fissures.*

3. How the Lungs stay "Attached" to the Thoracic Wall

- *The lungs themselves are directly covered by the visceral pleura, which sits immediately adjacent to the parietal pleura lining the thoracic wall.*

- *Between these two membranes is a thin, liquid-filled space called the pleural cavity, containing a few milliliters of intrapleural fluid.*

- *Surface Tension & Negative Pressure: The surface tension of this thin fluid film, combined with the negative intrapleural pressure, acts like a wet glass slide placed on a wet tabletop. This holds the visceral pleura (and thus the lungs) tightly against the parietal pleura (which lines the thoracic wall).*

- *When the thoracic wall expands during inspiration, it pulls the parietal pleura, which in turn pulls the visceral pleura and expands the lungs.*

Why the other options are incorrect:

- A. Pericardial membrane: Encloses the heart, not the lungs.
- C. Visceral pleural membrane: Directly covers the lung tissue itself, not the thoracic walls.
- D. Mesentery: A double layer of peritoneum that attaches abdominal organs to the posterior abdominal wall.
- E. Diaphragm: A major muscle of respiration forming the floor of the thoracic cavity, not the attachment membrane linking the lungs to the thoracic walls.

Which of the following hiatus is formed by muscular fibers arising from the right crus of the diaphragm?

- A. Caval hiatus
- B. Esophageal hiatus
- C. Aortic hiatus
- D. Phrenic hiatus

The correct answer is B. Esophageal hiatus.

The diaphragm has three major openings (hiatuses) through which vital structures pass between the thorax and the abdomen. The anatomical origin and level of each hiatus are high-yield board concepts:

1. The Esophageal Hiatus (Option B)

- Anatomical Level: T10

- **Formation:** It is located in the muscular part of the diaphragm and is formed by a sling of muscular fibers arising almost entirely from the RIGHT CRUS of the diaphragm.

- **Significance:** This muscular sling acts as a physiological sphincter for the esophagus. When the diaphragm contracts during inspiration, it pinches the esophagus, preventing gastroesophageal reflux.

- **Structures Passing Through:** Esophagus, Anterior and Posterior Vagal Trunks, and esophageal branches of the left gastric vessels.

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During phonation, which of the following structural changes will not necessarily happen?

A. Vocal folds adduct

B. Vestibular folds adduct

C. Rima glottidis close

D. Vestibular folds open

The correct answer is B. Vestibular folds adduct

To produce voice (phonation), the intrinsic muscles of the larynx adjust the position and tension of the true vocal folds (vocal cords):

1. What MUST happen during Phonation:

- **A. Vocal folds adduct:** The lateral cricoarytenoid and arytenoid muscles contract to pull the true vocal folds together into the midline (adduction).

- C. Rima glottidis closes: As the true vocal folds adduct, the space between them (the rima glottidis) narrows or closes completely. Air forced from the lungs through this narrowed slit causes the true vocal folds to vibrate, generating sound waves.

2. The Role of the Vestibular Folds (False Vocal Cords):

- The vestibular folds (false vocal cords) play no active role in normal phonation.
- Their primary anatomical job is protective—they adduct (close) during swallowing or heavy straining (Valsalva maneuver) to completely seal the airway and prevent aspiration.
- During normal speech and phonation, the vestibular folds remain open and relaxed to allow the sound waves produced by the true vocal folds to pass through freely without distortion.
- Therefore, adduction of the vestibular folds (Option B) does NOT happen during normal phonation (and if it does, it produces an abnormal, raspy voice known as dysphonia plicae ventricularis).

Which of the following paranasal air sinus drain into the ethmoidal bulla of middle nasal meatus?

- A. Frontal sinus
- B. Maxillary sinus
- C. Middle ethmoidal air sinus
- D. Anterior ethmoidal air sinus

Correct Answer: C. Middle ethmoidal air sinus

● Frontal sinus

Drainage: Middle nasal meatus → ethmoidal infundibulum

● Maxillary sinus

Drainage: Middle nasal meatus → hiatus semilunaris

● Anterior ethmoidal cells

Drainage: Middle nasal meatus → ethmoidal infundibulum

● Middle ethmoidal cells

Drainage: Middle nasal meatus → ethmoidal bulla

● Posterior ethmoidal cells

Drainage: Superior nasal meatus

● Sphenoidal sinus

Drainage: Sphenoethmoidal recess

Which layer is pierced last when a needle pass through the intercostal space of the chest wall into the pleural space?

- A. Parietal pleura
- B. Internal intercostal muscle
- C. Innermost intercostal muscle
- D. Endothoracic fascia

Correct Answer: A. Parietal pleura

Layers Pierced in Order

As a needle moves from the outside of the chest wall inward into the pleural cavity, it passes through these layers in order:

- ▼ *Skin and subcutaneous tissue*
- ▼ *External intercostal muscle*
- ▼ *Internal intercostal muscle*
- ▼ *Innermost intercostal muscle*
- ▼ *Endothoracic fascia*
- ▼ *Parietal pleura (pierced last before entering the pleural space).*

Which of the following is the correct arrangement of bronchopulmonary segments in the inferior lobe of left lung?

- A. Apico-posterior, anterior, superior linguar, inferior linguar*
- B. Superior, posterior basal, lateral basal, anterior basal*
- C. Apical, posterior, anterior*
- D. Middle lateral*

Correct Answer: B. Superior, posterior basal, lateral basal, anterior basal

Lung Lobe Segments

Left superior lobe contains the apicoposterior, anterior, and lingular segments.

Left inferior lobe contains the superior, medial basal, anterior basal, lateral basal, and posterior basal segments.

During quiet respiration, which ribs are crossed by the inferior margin of the lung in the midclavicular and the midaxillary lines, respectively?

- A. Ribs VI and VII*
- B. Ribs VI and VIII*
- C. Ribs II and VII*
- D. Ribs VIII and IX*

The correct answer is B. Ribs VI and VIII.

A fundamental, high-yield concept in surface anatomy is the "Rule of 2s" for comparing the surface extent of the Lungs versus the Pleural Reflections during quiet breathing.

1. The Surface Extent of the Lungs (The "6 - 8 - 10" Rule)

In quiet respiration, the lower border of the lungs crosses the thoracic wall at the following rib levels:

- Midclavicular Line: 6th Rib*
- Midaxillary Line: 8th Rib*
- Paravertebral / Scapular Line: 10th Rib*

Therefore, in the midclavicular and midaxillary lines, the inferior margin of the lung crosses Rib VI and Rib VIII, respectively.

2. The Surface Extent of the Pleura (The "8 - 10 - 12" Rule)

For comparison, the parietal pleural reflection extends two ribs lower than the lung margin:

- Midclavicular Line: 8th Rib
- Midaxillary Line: 10th Rib
- Paravertebral / Scapular Line: 12th Rib

The inferior demifacet of the 4th thoracic vertebra receives:

- A. Superior articular facet of the 4th rib
- B. Inferior articular facet of the 4th rib
- C. Superior articular facet of the 5th rib
- D. Inferior articular facet of the 5th rib

Correct Answer: C. Superior articular facet of the 5th rib

For a typical thoracic vertebra:

Superior demifacet ↔ inferior facet of the same-numbered rib

Inferior demifacet ↔ superior facet of the next-numbered rib

If the question says:

"Inferior demifacet of T6 articulates with..."

Immediately think:

Inferior → next rib

⇒ Superior facet of rib 7 

If it says:

"Superior demifacet of T6..."

Think:

Superior → same rib

⇒ Inferior facet of rib 6 

A large collection of lymphoid tissue, the pharyngeal tonsil, lies in the mucosa covering the roof of the;

A. Oropharynx

B. Nasopharynx

C. Laryngopharynx

D. Trachea

Correct Answer: B. Nasopharynx

Which of the following anatomical structures are not structures in the "conducting zone" of the respiratory tract?

A. Nasal cavity, larynx and trachea.

B. Primary, secondary and tertiary bronchi and bronchioles.

C. Nares, conchae, olfactory mucosa and sinuses.

D. Alveolar ducts, alveolar sacs, and alveoli.

Correct Answer: D. Alveolar ducts, alveolar sacs, and alveoli

Key Distinction

<i>Conducting zone</i>	<i>Respiratory zone</i>
<i>Nasal cavity</i>	<i>Respiratory bronchioles</i>
<i>Pharynx</i>	<i>Alveolar ducts</i>
<i>Larynx</i>	<i>Alveolar sacs</i>
<i>Trachea</i>	<i>Alveoli</i>
<i>Primary bronchi</i>	
<i>Secondary bronchi</i>	
<i>Tertiary bronchi</i>	
<i>Bronchioles</i>	
<i>Terminal bronchioles</i>	

The term exactly defines to the gases exchange between air in alveolar sacs and blood in lungs?

- A. Internal respiration*
- B. External respiration*
- C. Ventilation*
- D. Cellular respiration*

The correct answer option is B. External respiration.

Why the correct option is right:

B. External respiration: This process explicitly refers to the exchange of oxygen and carbon dioxide across the respiratory membrane between the air in the alveoli (air sacs) and the blood

flowing through the pulmonary capillaries in the lungs.

Why the other options are wrong:

✗ A. Internal respiration: This describes the exchange of gases at the tissue level, moving oxygen from the systemic capillaries into body cells and carbon dioxide out of the cells into the blood.

✗ C. Ventilation: This is simply the mechanical process of breathing (inhalation and exhalation) which moves bulk air into and out of the lungs, but it does not describe the microscopic gas exchange itself.

✗ D. Cellular respiration: This is an intracellular metabolic pathway where cells break down glucose using oxygen to generate ATP energy, producing carbon dioxide as a chemical waste product.

Which of the following two anatomical structures of the respiratory system are connected with Larynx?

A. The nares and the choanae

B. The pharynx and the trachea

C. The choanae and the glottis

D. The trachea and choanae

Correct Answer: B. The pharynx and the trachea

Why the other options are wrong

A. Nares and choanae ✗

These are structures associated with the nasal cavity.

Nares = openings of the nose

Choanae = posterior openings of nasal cavities into the nasopharynx

They do not directly connect to the larynx.

C. Choanae and glottis ❌

The glottis is a structure within the larynx, not a separate structure connected to it.

D. Trachea and choanae ❌

The trachea connects inferiorly with the larynx, while the choanae are openings between the nasal cavity and nasopharynx.

The walls of the following respiratory structures are all supported by cartilage EXCEPT:

A. Bronchioles

B. Trachea

C. Bronchi

D. Larynx

The correct answer option is A. Bronchioles.

Why the correct option is right

● A. Bronchioles: By biological definition, bronchioles are small respiratory airways that completely lack cartilaginous support in their walls. Instead, their structural integrity and diameter are regulated entirely by a thick layer of smooth muscle and elastic fibers. This allows them to constrict or dilate to control airflow volume.

Why the other options are wrong

● B. Trachea: The trachea (windpipe) is heavily reinforced by 16 to 20 C-shaped rings of

hyaline cartilage that prevent the airway from collapsing during inhalation.

● C. Bronchi: The primary, secondary, and tertiary bronchi are all structurally supported by rings or shifting plates of cartilage to maintain an open airway as they enter the lungs.

● D. Larynx: The larynx (voice box) is a complex framework composed entirely of multiple pieces of cartilage, including the thyroid, cricoid, and epiglottic cartilages.

Which of the following anatomical structures constitute only "anatomical structures of upper respiratory tract"?

A. Nose, pharynx and larynx

B. Larynx, epiglottis and bronchi

C. Trachea, bronchi and bronchioles

D. Terminal bronchioles, alveoli and pleurae

A. Nose, pharynx and larynx constitute only the upper respiratory tract. The upper tract includes structures outside the chest or above the vocal cords, while options B, C, and D contain lower respiratory structures like the bronchi, bronchioles, and alveoli.

Respiratory Tract Breakdown

Upper Respiratory Tract

▼ Nose: Filters and warms air.

▼ Pharynx: Throat passage for air and food.

▼ Larynx: Voice box protecting the lower airway.

Lower Respiratory Tract

- ▼ Trachea: Windpipe leading to the lungs.
- ▼ Bronchi: Main airways branching into lungs.
- ▼ Bronchioles: Smaller branches inside the lungs.
- ▼ Alveoli: Tiny air sacs for gas exchange.

Which part of the parietal pleura is different in terms of its innervation?

- A. Costal pleura
- B. Mediastinal pleura
- C. Most of diaphragmatic part
- D. Cervical pleura

The correct answer is B. Mediastinal pleura (along with the central part of the diaphragmatic pleura).

To understand why this is a classic, we must look at the dual somatic innervation of the parietal pleura:

1. Parietal Pleura Innervated by the Phrenic Nerve (C3, C4, C5)

- Mediastinal Pleura (lining the lateral sides of the pericardium/mediastinum).
- Central Part of Diaphragmatic Pleura.
- Clinical Effect: Pain originating from these regions travels via the phrenic nerve and is referred to the shoulder tip / base of the neck (C3-C5 dermatomes).

2. Parietal Pleura Innervated by Intercostal Nerves (T1-T11)

- Costal Pleura (lining the inside of the ribs and intercostal spaces).*
- Cervical Pleura (dome of the pleura over the apex of the lung).*
- Peripheral / Most of Diaphragmatic Part (outer margins of the diaphragm).*
- Clinical Effect: Pain originating from these regions is localized directly to the chest wall or upper abdominal wall corresponding to the intercostal dermatomes.*

Why Mediastinal Pleura (Option B) stands out:

- Options A (Costal), C (Most of diaphragmatic / peripheral), and D (Cervical) are all supplied by intercostal nerves (somatic spinal nerves T1-T11).*
- Option B (Mediastinal pleura) is the only major parietal pleura region in this list that is innervated by a completely different nerve source—the Phrenic Nerve (C3-C5).*

Anatomically speaking if small foreign body enter into the trachea, then more probably it will enter into:-

- A. Upper lobar bronchus of left lung*
- B. Lower lobar bronchus of left lung*
- C. Upper lobar bronchus of right lung*
- D. Lower lobar bronchus of right lung*

Correct Answer: D. Lower lobar bronchus of right lung

Which of the following statements about typical left lung is true?

- A. It is divided by horizontal and oblique fissures into a superior, middle, and inferior lobe.*
- B. Is short and broader than the right*
- C. The superior vena cava is intimately associated with its mediastinal surface, where it makes a groove.*
- D. Its anterior border has a characteristic cardiac notch*

The correct answer is D. Its anterior border has a characteristic cardiac notch.

Why D is correct:

- Because the heart deviates significantly to the left side of the thoracic cavity, the anterior border of the left lung features a prominent concavity called the cardiac notch.*
- Below this notch, the anterior border projects forward as a tongue-like process called the lingula (which is anatomically homologous to the middle lobe of the right lung).*

Why the other options are incorrect:

- A is incorrect: The left lung is divided into only two lobes (superior and inferior) by a single oblique fissure. It does not have a middle lobe or a horizontal fissure. (The right lung is the one divided into three lobes by both an oblique and a horizontal fissure).*

• B is incorrect: The left lung is longer and narrower than the right lung (because the heart occupies space on the left, compressing its width, while the right lung is shorter and broader because the liver pushes the right dome of the diaphragm upward).

• C is incorrect: The Superior Vena Cava (SVC) leaves a groove on the mediastinal surface of the RIGHT lung, not the left lung. The mediastinal surface of the left lung features prominent grooves for the Arch of the Aorta, the Descending Thoracic Aorta, and the Left Subclavian Artery.

In the contents of lung roots the most superior structure at the hilum of left lung is?

A. Primary bronchus

B. Superior pulmonary vein

C. Pulmonary artery

D. Inferior pulmonary vein

The correct answer option is C. Pulmonary artery.

Explanation

Within the root and hilum of the left lung, the structures follow a specific superior-to-inferior and anterior-to-posterior arrangement:

● C. Pulmonary artery: This is the most superior structure at the left hilum. It arches over the left main bronchus as it carries deoxygenated blood from the heart to the left lung.

✗ A. Primary bronchus: The left primary (main) bronchus sits posteroinferior (below and behind) relative to the pulmonary artery.

✗ B. Superior pulmonary vein: This vessel lies anterior and slightly inferior to the pulmonary

artery, making it the most anterior structure rather than the most superior.

✗ D. Inferior pulmonary vein: This structure is located at the lowest (most inferior) portion of the left hilum.

Which of the following two anatomical structures involve in both in air transportation and gas exchange?

A. Respiratory bronchioles

B. Terminal bronchioles

C. Segmental bronchi

D. Primary bronchus

Correct Answer: A. Respiratory bronchioles

A 46 years old woman was admitted for headaches and impaired vision, and MRI and CT revealed a pituitary adenomas. The physician decided to remove the cancer. Pituitary can be accessed surgically by passing instruments through the;

A. Sphenoid bone and sinus

B. Maxillary bone and maxillary sinus

C. Frontal bone and frontal sinus

D. Ethmoid bone and sinus

E. None

The correct answer is A. Sphenoid bone and sinus.

☀ Anatomical Proximity: The pituitary gland resides within a bony depression called the sella turcica, which is located directly above and behind the sphenoid sinus.

 **Standard Surgical Approach:** The primary minimally invasive procedure to remove pituitary tumors is called transsphenoidal surgery. During this approach, neurosurgeons advance instruments directly through the nasal cavity and the sphenoid bone/sinus to reach the tumor safely without performing a traditional craniotomy.

Regarding the pleura the incorrect statement is:

- A. Cervical part is above the clavicle
- B. Diaphragmatic pleura supplied by intercostal nerves only
- C. Cervical pleura is crossed by subclavian vessels –
- D. Pleural cavity is a potential space
- E. The visceral & parietal pleurae are continuous around the root of the lung

B. Diaphragmatic pleura supplied by intercostal nerves only.

Explanation of the Incorrect Statement

 Statement B is incorrect because the diaphragmatic parietal pleura has a dual nerve supply. The peripheral part of the diaphragmatic pleura is innervated by the lower intercostal nerves (T7–T12), while the central part (covering the central tendon of the diaphragm) is innervated by the phrenic nerves (C3–C5). Therefore, saying it is supplied by intercostal nerves only is false.

Why the Other Statements Are Correct

 A. Cervical part is above the clavicle: The cervical pleura (or cupula) extends upwards into the root of the neck, projecting roughly 2–3 cm above the level of the medial third of the clavicle.

 C. Cervical pleura is crossed by subclavian vessels: As the subclavian artery and vein arch over the first rib to reach the upper limb, they directly cross anterior to the cervical pleura.

♥ D. Pleural cavity is a potential space: Under normal physiological conditions, the space between the visceral and parietal layers contains only a very thin layer of serous fluid to facilitate smooth sliding, meaning it is a closed "potential" space rather than an actual open void.

♥ E. The visceral & parietal pleurae are continuous around the root of the lung: At the hilum (root) of each lung, the parietal pleura reflects off the mediastinum to become the visceral pleura, creating a continuous membranous connection.

A medical practitioner places a stethoscope over the patient's seventh right intercostal space in the mid-axillary line. The stethoscope overlies the;

- A. Upper lobe of the lung.
- B. Middle lobe of the lung.
- C. Lower lobe of the lung.
- D. Costo-diaphragmatic recess.
- E. None

The correct answer is C. Lower lobe of the lung.

To solve this, we must map out the surface anatomy of the right lung's fissures and lobes relative to the 7th intercostal space at the mid-axillary line.

1. Fissure Mapping on the Right Lung:

- Horizontal Fissure: Follows the course of the 4th rib anteriorly to intersect the oblique fissure in the mid-axillary line at the level of the 5th rib.
- Oblique Fissure: Begins posteriorly at the spinous process of T3/T4, crosses the 5th rib in the mid-axillary line, and continues forward to reach the 6th costal cartilage anteriorly.

2. Lobe Territories in the Mid-Axillary Line:

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Above the Oblique Fissure (Above the 5th rib):

- Upper Lobe: Above the 4th rib.
- Middle Lobe: Between the 4th and 5th ribs anteriorly/laterally.

•

Below the Oblique Fissure (Below the 5th rib):

- The Lower Lobe occupies the entire lower lateral and posterior space beneath the oblique fissure (from the 5th rib down to the 8th rib in the mid-axillary line during quiet respiration).

3. Placement at the 7th Intercostal Space (Mid-Axillary Line):

- The 7th intercostal space lies between the 7th and 8th ribs.
- This is well below the oblique fissure (5th rib) and above the inferior lung margin (8th rib in the mid-axillary line).
- Therefore, placing a stethoscope over the 7th right intercostal space in the mid-axillary line directly overlies the lower lobe of the right lung.

Why the other options are incorrect:

- A. Upper lobe: Located superior to the 4th rib.
- B. Middle lobe: Located between the 4th rib and 5th rib anteriorly/laterally.
- D. Costo-diaphragmatic recess: In the mid-axillary line, the costodiaphragmatic recess spans between the 8th and 10th ribs. At the 7th intercostal space, lung tissue (lower lobe) is still actively present inside the pleural cavity during respiration.

Which of the following paranasal sinuses open into the middle meatus?

- A. The anterior ethmoidal sinuses
- B. The anterior ethmoidal and frontal sinuses
- C. The anterior ethmoidal, frontal and maxillary sinuses
- D. The anterior ethmoidal, frontal, maxillary and sphenoidal sinuses

Correct Answer: C

The middle nasal meatus receives drainage from:

- ☒ Frontal sinus
- ☒ Maxillary sinus
- ☒ Anterior ethmoidal air cells
- ☒ Middle ethmoidal air cells

Therefore, among the options, C is the best answer.

◆ Understand the Drainage Pathways

The middle meatus contains two particularly important structures:

1 □ Ethmoidal infundibulum

Receives drainage from:

● Frontal sinus

● Anterior ethmoidal air cells

2 □ Hiatus semilunaris

Receives drainage from:

● Maxillary sinus

● Anterior ethmoidal cells via the infundibulum

3 □ Ethmoidal bulla

Receives drainage from:

● Middle ethmoidal air cells

The vertebrae are interlocked via their:

A. Articular processes

B. Spinous processes

C. Transverse processes

D. Body

Correct Answer: A. Articular processes

▼ *Articular processes: Fit together to form facet joints. These joints lock the bones in place and guide movement.*

▼ *Spinous processes: Point out at the back. They give muscles a place to attach, but they do not lock the bones together.*

▼ *Transverse processes: Stick out on the sides. They also help attach muscles and ribs, but do not join bones.*

▼ *Body: The large front part of the bone. It bears weight, separated by soft discs, but it does not interlock directly.*

Flail chest is defined as:

A. Multiple rib fractures with subsequent subcutaneous emphysema

B. Chyle in the pleural space

C. Excess fluid in pericardium

D. Two or more ribs fractured at two points

The correct answer is D. Two or more ribs fractured at two points (or three or more contiguous ribs fractured at two points each).

Flail chest is a severe, life-threatening thoracic trauma condition that impairs the normal mechanics of respiration.

1. Definition (Option D)

- Anatomically, flail chest occurs when two or more contiguous (adjacent) ribs are fractured in two or more places.

- This creates a free-floating, unstable segment of the thoracic chest wall that is physically detached from the rest of the rib cage.

2. The Hallmark Feature: Paradoxical Respiration

Because the flail segment is detached from the rigid chest cage, it loses structural stability and responds passively to changes in intra-thoracic pressure during breathing:

- **Inspiration:** Normally, the chest wall expands outward, creating negative intra-thoracic pressure to pull air in. In flail chest, this negative pressure sucks the unstable flail segment **INWARD**.

- **Expiration:** Normally, the chest wall moves inward as air leaves. The positive intra-thoracic pressure pushes the unstable flail segment **OUTWARD**.

This abnormal, counter-intuitive movement is called paradoxical chest movement and leads to severe hypoxemia, respiratory fatigue, and underlying pulmonary contusion (lung bruising).

Why the other options describe completely different conditions:

- **A. Multiple rib fractures with subsequent subcutaneous emphysema:** While subcutaneous emphysema (air under the skin) can occur alongside severe rib fractures, this does not define the structural, mechanical instability of a flail chest segment.

- **B. Chyle in the pleural space:** This is the definition of a Chylothorax (caused by rupture or obstruction of the thoracic duct).

- C. Excess fluid in pericardium: This is the definition of a Pericardial Effusion (which can lead to Cardiac Tamponade).

Which structure below leaves the most pronounced impression on the right lung of the cadaver?

- A. Right vagus nerve
- B. Azygos vein
- C. Right phrenic nerve
- D. Thoracic aorta
- E. Right common carotid artery

The correct answer is B. Azygos vein.

When examining the mediastinal surface of a preserved (cadaveric) right lung, several structures leave distinct grooves and impressions. However, the arch and pathway of the azygos vein form the most prominent and characteristic venous impression on the right side.

1. The Impression of the Azygos Vein (Option B)

- The azygos vein ascends in the posterior mediastinum along the right side of the vertebral column.
- At the level of the T4/T5 vertebra (above the root of the right lung), it arches forward over the superior aspect of the right main bronchus and pulmonary vessels to empty into the Superior Vena Cava (SVC).
- This creates a deep, bold, C-shaped groove called the Groove for the Arch of the Azygos Vein, wrapping directly superior to the hilum of the right lung.

Why the other options are incorrect or leave less/no impression:

- D. Thoracic aorta: Leaves a massive, prominent groove on the LEFT lung, not the right lung. (The arch of the aorta and descending thoracic aorta run on the left side of the posterior mediastinum).
- A. Right vagus nerve & C. Right phrenic nerve: Nerves are thin, soft, string-like structures. They pass over the lung surface, but they are far too small and pliable to leave a deep, "pronounced" bony/venous impression on the lung tissue during cadaveric fixation.
- E. Right common carotid artery: Arises from the brachiocephalic artery behind the right sternoclavicular joint and passes up into the neck, leaving only a minor, faint groove near the apex of the right lung (the brachiocephalic vein/artery groove), which is far less pronounced than the major azygos arch.

During inspiration, air passes into lungs due to;

- A. Increase in volume of thoracic cavity and fall in lung pressure
- B. Fall in pressure inside the lungs
- C. Increased volume of thoracic cavity
- D. Muscular expansion of lungs

Correct Answer: A. Increase in volume of thoracic cavity and fall in lung pressure

Which of the following structures/features are likely to be located at the level of the sternal angle:

- A. Right lobar bronchus
- B. Left recurrent laryngeal nerve
- C. Left brachiocephalic vein

D. Carinal ridge

E. Base of the ascending aorta

Correct Answer: D. Carinal ridge

◆ Major Structures at the Sternal Angle

The sternal angle is a major anatomical landmark because several important structures occur at this level:

🫁 Respiratory

Trachea → bifurcates into right and left main bronchi

→ Carina ★

💧 Cardiovascular

The ascending aorta becomes the arch of the aorta around this level.

The arch of the aorta begins and ends approximately in this transverse plane.

💧 Great veins

The left brachiocephalic vein joins the right brachiocephalic vein to form the superior vena cava more inferiorly, so the left brachiocephalic vein itself is not the classic landmark answer here.

🫀 Aortic arch

The transverse plane is also associated with the:

Beginning of the aortic arch

End of the aortic arch

Why the Other Options Are Wrong?

A. Right lobar bronchus ❌

The trachea divides into the right and left main bronchi at the sternal angle.

The right lobar bronchi arise further distally.

B. Left recurrent laryngeal nerve ❌

The left recurrent laryngeal nerve loops under the arch of the aorta, near the ligamentum arteriosum.

Its relationship is important, but it is not the structure classically identified with the sternal angle.

C. Left brachiocephalic vein ❌

The left brachiocephalic vein is formed by the union of the left internal jugular and left subclavian veins and passes obliquely across the superior mediastinum.

It is not the classic structure at the sternal angle.

E. Base of the ascending aorta ❌

The ascending aorta begins at the left ventricle and lies inferior to the sternal-angle plane.

The important aortic landmark at the sternal angle is the transition from ascending aorta to aortic arch.

All of the following are characteristics of bronchopulmonary segments except?

A. It is named according to the segmental bronchus supplying it

B. It is surgically respectable

C. It is largest division of lobe

D. It is drained by intra segmental branch of pulmonary vein

The correct answer is D. It is drained by intra segmental branch of pulmonary vein.

This question tests a fundamental, high-yield concept in surgical lung anatomy: the relationship of bronchi, arteries, and veins to a bronchopulmonary segment.

Why Option D is INCORRECT (and therefore the correct answer to "except"):

- The Pulmonary Vein Pathway: The branches of the pulmonary veins do not run through the center of a bronchopulmonary segment (they are NOT intrasegmental).

- Instead, pulmonary veins run in the connective tissue septa BETWEEN adjacent segments—they are INTERsegmental.

- Surgical Significance: Because the pulmonary veins are intersegmental, they act as anatomical boundaries during surgery and drain blood from the margins of adjacent segments.

Conversely, the segmental bronchus and segmental branch of the pulmonary artery run together right through the CENTER (intrasegmental) of the segment.

Why the other options ARE true characteristics of Bronchopulmonary Segments:

- A. It is named according to the segmental bronchus supplying it: TRUE. Each segment is a pyramid-shaped anatomical unit supplied by a specific tertiary (segmental) bronchus (e.g., Apical, Posterior, Anterior segments).

- B. It is surgically resectable: TRUE. Because each segment has its own isolated bronchus, arterial supply, and fibrous tissue boundaries, a surgeon can cleanly excise a diseased segment (segmentectomy) without damaging neighboring healthy segments.

- C. It is the largest division of a lobe: TRUE. The anatomical hierarchy of the lung is: Whole Lung → Lobes → Bronchopulmonary Segments (largest subdivision of a lobe) → Pulmonary Lobules.

Exam: Gross Anatomy

Year: 2016

Dr. Doc Tutorial

The superior articular facet of rib 5 articulates with the;

- A. Spinous process of the corresponding vertebra
- B. Superior costal facet of the corresponding vertebra
- C. Inferior costal facet of T4
- D. Transverse costal facet of T5
- E. Body of T5 vertebra

C. Inferior costal facet of T4

Explanation: A typical rib (Ribs 3-9) articulates via its head with two vertebral bodies. The superior facet articulates with the inferior costal facet of the vertebra above (T4), while the inferior facet articulates with the superior costal facet of its own vertebra (T5).

The lungs are attached to the walls of the thorax by means of the:

- A. Pericardial membrane
- B. Parietal pleural membrane
- C. Visceral pleural membrane
- D. Mesentery
- E. Diaphragm

B. Parietal pleural membrane

Explanation: The parietal pleura lines the internal surface of the thoracic wall. Surface tension from the thin fluid layer between the parietal and visceral pleurae holds the lungs attached and expanded against the chest wall.

Which of the following intrinsic muscles is not responsible for the adductor of vocal cords?

- A. Lateral crico-arytenoids
- B. Posterior crico-arytenoids
- C. Oblique arytenoids
- D. Transverse arytenoids
- E. Thyroarytenoids

B. Posterior crico-arytenoids

Explanation: The posterior cricoarytenoid is the ONLY abductor (opener) of the vocal cords. All other intrinsic laryngeal muscles listed act as adductors (closers) or tensors.

The left external branch of superior laryngeal nerve injury (superior laryngeal neuralgia)

during a thyroid surgery procedure could possibly result in;

- A. Bilateral vocal paralysis
- B. Unilateral loss of pitch elevation and vocal fatigue
- C. Loss of sensory to mucous membrane above vocal fold
- D. Loss of sensory to mucous membrane below vocal fold

The correct answer is B. Unilateral loss of pitch elevation and vocal fatigue.

The Superior Laryngeal Nerve (SLN) (a branch of the Vagus nerve, CN X) divides into two main branches near the upper pole of the thyroid gland:

1. External Branch of SLN (Motor):

Course: Runs in close proximity to the superior thyroid artery (making it vulnerable to accidental injury during thyroidectomy when the artery is ligated).

Innervation: Provides motor innervation exclusively to the cricothyroid muscle.

Function of Cricothyroid: Tenses and elongates the vocal cords to elevate vocal pitch (producing high-frequency sounds).

*** Effect of Injury:** Paralysis of the cricothyroid muscle leads to unilateral loss of high-pitched voice, a weak/monotonous voice, and rapid vocal fatigue.

2. Internal Branch of SLN (Sensory):

** Function: Pierces the thyrohyoid membrane to provide sensory innervation to the laryngeal mucosa above the vocal folds (including the epiglottis and vestibule).*

** Injury here causes Option C.*

Why the other options are incorrect:

A. Bilateral vocal paralysis: Caused by bilateral injury to the Recurrent Laryngeal Nerves (RLN), leading to airway obstruction and aphonia.

C. Loss of sensory to mucous membrane above vocal fold: Caused by injury to the internal branch of the superior laryngeal nerve, not the external branch.

D. Loss of sensory to mucous membrane below vocal fold: Caused by injury to the Recurrent Laryngeal Nerve (RLN), which provides sensory innervation to the larynx below the level of the true vocal folds.

A large hilar lymph node is found over the most superior aspect of the hilum of the left lung in a primary tuberculosis patient. Which of the following structures in the root of the specified lung is most likely to be compressed?

A. Pulmonary artery

B. Main bronchus

C. Hyparterial bronchus

D. Pulmonary vein

E. Pulmonary plexus

A. Pulmonary artery

Explanation: In the root/hilum of the left lung, the pulmonary artery is the most superior structure ("Artery-Bronchus-Vein" from superior to inferior).

The largest and the most infected paranasal air sinus is the;

A. Maxillary sinus

B. Sphenoid sinus

C. Middle ethmoid sinus

D. Frontal sinus

A. Maxillary sinus

Explanation: The maxillary sinus is the largest paranasal sinus and the most frequently infected because its drainage ostium is located high on its medial wall, preventing easy gravity drainage.

The lateral & medial bronchopulmonary segments belong to;

A. Superior lobe of left lung

B. Middle lobe of right lung

C. Inferior lobe of right lung

D. Superior lobe of right lung

B. Middle lobe of right lung

Explanation: The middle lobe of the right lung is divided into two bronchopulmonary segments: the lateral (Segment IV) and medial (Segment V) segments.

Which structure below leaves the most pronounced impression on mediastinal surface of the right lung?

- A. Right vagus nerve
- B. Azygos vein
- C. Esophagus
- D. Thoracic aorta
- E. Right common carotid artery

B. Azygos vein

Explanation: The azygos vein arches over the root of the right lung to empty into the Superior Vena Cava, leaving a deep, prominent C-shaped groove on the right lung's mediastinal surface.

The stimulation of bronchial glands secretion, contraction of the bronchial smooth muscle, and vasodilation of the pulmonary vessels is stimulated by the;

- A. Sympathetic from cervical sympathetic plexus
- B. Parasympathetic from vagal trunk
- C. Visceral afferent from sensory ganglion of the vagus nerve
- D. Parasympathetic glossopharyngeal nerve

B. Parasympathetic from vagal trunk

Explanation: Parasympathetic fibers (via the Vagus nerve, CN X) cause bronchoconstriction, increased bronchial gland secretion, and vasodilation of pulmonary vessels. (Sympathetic causes bronchodilation and vasoconstriction).

A medical practitioner places a stethoscope over the patient's seventh right intercostal space in the mid-axillary line. The stethoscope overlies the;

- A. Upper lobe of the lung
- B. Middle lobe of the lung
- C. Lower lobe of the lung
- D. Costo-diaphragmatic recess
- E. None of the above

C. Lower lobe of the lung

• Explanation: At the 7th intercostal space in the mid-axillary line, the stethoscope overlies the lower lobe of the right lung, which sits beneath the oblique fissure (5th rib level).

Which structure does the central tendon of the diaphragm fuse with?

- A. Pericardium
- B. Greater omentum
- C. Intercostal muscles
- D. Ribs 7-12
- E. Parietal pleurae

A. Pericardium

• Explanation: The central tendon of the diaphragm is directly fused superiorly with the fibrous pericardium of the heart.

Which of the following is true about the course of the intercostal nerves?

- A. They run between the parietal and visceral pleura.
- B. They run between the internal and external intercostal muscles.

C. They run along the superior border of the rib below.

D. They run along the inferior border of the rib above.

D. They run along the inferior border of the rib above.

• Explanation: The intercostal neurovascular bundle (VAN: Vein, Artery, Nerve) runs in the costal groove along the inferior border of the rib above, between the internal and innermost intercostal muscles.

Which of the following is not a function of the paranasal sinuses?

A. Producing mucus

B. Filtering air

C. Lightening the weight of the skull

D. Regulating temperature

B. Filtering air

• Explanation: Air filtration (via vibrissae and mucociliary clearance) occurs in the nasal cavity proper.

All of the following are characteristics of bronchopulmonary segments except?

A. It is named according to the segmental bronchus supplying it

B. It is surgically respectable

C. It is largest division of lobe

D. It is drained by intra segmental branch of pulmonary vein

D. It is drained by intra segmental branch of pulmonary vein

• Explanation: Pulmonary veins run INTERsegmentally (in the connective tissue boundaries between segments). The segmental bronchus and pulmonary artery run INTRAsegmentally (down the center).

Which of the following muscles is responsible for normal expiration?

- A. Diaphragm
- B. External intercostals
- C. Internal intercostals
- D. None of the above

D. None of the above

• Explanation: Normal (quiet) expiration is a completely passive process driven by the elastic recoil of the lungs and chest wall. No muscular contraction is required.

Which structure is located immediately anterior to the right lung hilum?

- A. Superior vena cava
- B. Aorta
- C. Esophagus
- D. Inferior vena cava

The correct answer is A. Superior vena cava.

When examining the mediastinal structures in relation to the right lung hilum (the root of the right lung), it's crucial to visualize their precise anterior-posterior and superior-inferior

relationships.

Structures Immediately Anterior to the Right Lung Hilum:

- *The Superior Vena Cava (SVC): This large vein runs vertically downwards along the right side of the superior mediastinum, immediately anterior to the right lung hilum, as it descends to enter the right atrium.*
- *The Phrenic Nerve: The right phrenic nerve also descends immediately anterior to the right lung hilum (and then anterior to the root of the lung).*

Structures Superior to the Right Lung Hilum:

- *The Arch of the Azygos Vein: Arches superiorly over the root of the right lung to drain into the SVC.*

Structures Posterior to the Right Lung Hilum:

- *The Esophagus*
- *The Azygos Vein (main ascending trunk)*
- *The Vagus Nerve (right vagus nerve)*

Why the other options are incorrect:

- *B. Aorta:* The aorta (specifically the ascending aorta and arch) is located more anterior and central in the mediastinum relative to the right lung hilum, but the SVC is literally immediately adjacent and anterior. The descending aorta is located posterior to the left lung hilum.

- *C. Esophagus:* The esophagus lies primarily posterior to the right lung hilum.

- *D. Inferior vena cava:* The IVC enters the right atrium from below (inferiorly) and is not immediately anterior to the lung hilum itself.

Which of the following is false about the articular structures of the thoracic typical ribs?

A. Head of the rib attaches with two adjacent vertebral bodies

B. Larger, superior costal facet for articulation with the vertebral body of its own number

C. Larger, inferior costal facet for articulation with the vertebral body of its own number

D. The tubercle of the rib articulates with the transverse process of vertebra

B. Larger, superior costal facet for articulation with the vertebral body of its own number

- *Explanation:* This statement is FALSE. The inferior costal facet is larger and articulates with its own vertebra; the superior facet is smaller and articulates with the vertebra above.

When excess fluid accumulates within the pleural space and may collapse the lung as the volume increases. This condition is called?

A. Pleural recess

B. Pleural effusion

C. Pleural aspiration

D. Pneumothorax

B. Pleural effusion

- *Explanation: Accumulation of excess fluid in the pleural space is called a pleural effusion. (Accumulation of air is a pneumothorax).*

The posterior mediastinum does not include which of the following?

- A. Thoracic duct*
- B. Thoracic sympathetic trunks*
- C. Phrenic nerves*
- D. Inferior vena cava*

C. Phrenic nerves

- *Explanation: The phrenic nerves run through the Middle Mediastinum (embedded in the fibrous pericardium). The posterior mediastinum contains the thoracic aorta, esophagus, thoracic duct, azygos veins, and sympathetic trunks.*

One of the following is true about the trachea

- A. Has continuous circular cartilage*
- B. Lined by stratified columnar epithelium*
- C. Bifurcates at the level of sternal angle into primary bronchi*
- D. None*

C. Bifurcates at the level of sternal angle into primary bronchi

• Explanation: The trachea bifurcates at the sternal angle (T4/T5 level). It has incomplete C-shaped cartilage rings (not continuous) and is lined by pseudostratified ciliated columnar epithelium.

One of the following muscles do not elevate the ribs

- A. External intercostal
- B. Serratus posterior superior
- C. Serratus posterior inferior
- D. None

C. Serratus posterior inferior

• Explanation: The serratus posterior inferior depresses the lower ribs (assists in expiration). External intercostals and serratus posterior superior elevate the ribs during inspiration.

Exam: Gross Anatomy

Year: 2017

Dr. Doc Tutorial

1. The segmental classification of the lung is used for:

- A. Anatomical description
- B. Surgical planning
- C. Functional analysis
- D. All of the above

D. All of the above

• Explanation: Bronchopulmonary segments are individual, functional, and anatomical units of the lung. They are essential for anatomical description, surgical planning (segmentectomy without damaging adjacent tissue), and functional/radiological analysis (localization of disease/ventilation-perfusion).

2. The last conducting zone that is purely conductive (before the respiratory zone begins) is the:

- A. Primary bronchus
- B. Respiratory bronchiole
- C. Alveolar duct
- D. Terminal bronchiole

D. Terminal bronchiole

• Explanation: The terminal bronchiole is the last and smallest structure of the conducting zone. Beyond this point, the respiratory zone begins with the respiratory bronchioles (which have alveoli in their walls for gas exchange).

3. The artery that supplies the supraepiglottic region is the:

- A. Inferior laryngeal artery
- B. Superior laryngeal artery
- C. Cricothyroid artery
- D. Lingual artery

The correct answer is B. Superior laryngeal artery.

The arterial supply of the larynx is cleanly divided into two regions by the vocal folds (glottis):

1. Superior Laryngeal Artery (Option B):

* Origin: Branch of the superior thyroid artery (from the External Carotid Artery).

* Course: Accompanies the internal branch of the superior laryngeal nerve and pierces the thyrohyoid membrane.

* Territory: Supplies the supraglottic / supraepiglottic region (everything above the level of the true vocal folds, including the epiglottis, aryepiglottic folds, and upper laryngeal mucosa).

2. Inferior Laryngeal Artery (Option A):

* Origin: Branch of the inferior thyroid artery (from the Thyrocervical trunk of the Subclavian Artery).

* Course: Accompanies the recurrent laryngeal nerve into the larynx.

* Territory: Supplies the infraglottic region (everything below the level of the true vocal folds).

4. The structure that is found immediately posterior to the larynx is the:

A. C3-C6 cervical vertebrae

B. Esophagus

C. Laryngotracheal tube

D. Oropharynx

B. Esophagus

• Explanation: The laryngopharynx / esophagus lies directly behind (posterior to) the larynx and trachea. (Note: The C3–C6 vertebrae lie posterior to the pharynx).

5. Regarding the diaphragm, which statement is TRUE?

A. The medial arcuate ligament arches over the quadratus lumborum muscle.

B. Its motor nerve supply is only by the phrenic nerve (C3, C4, C5).

C. The esophagus pierces the diaphragm at the T-8 vertebral level.

D. The vena caval opening transmits the IVC and the left phrenic nerve.

B. Its motor nerve supply is only by the phrenic nerve (C3, C4, C5).

• Explanation:

• B is TRUE: The phrenic nerve provides 100% of the motor supply to the entire diaphragm.

• Why others are false: Medial arcuate ligament arches over psoas major (lateral arcuate over quadratus lumborum); Esophagus pierces at T10; Caval hiatus transmits IVC and RIGHT phrenic nerve.

6. The Left recurrent laryngeal nerve loops around the:

A. Left subclavian artery

B. Brachiocephalic trunk

C. Arch of the aorta

D. Clavicle

C. Arch of the aorta

• Explanation: The left recurrent laryngeal nerve branches off the vagus nerve in the superior mediastinum and loops under the arch of the aorta (just lateral to the ligamentum arteriosum) before ascending in the tracheoesophageal groove. (The right recurrent laryngeal nerve loops around the right subclavian artery).

7. The nerve that supplies all intrinsic laryngeal muscles EXCEPT the cricothyroid muscle is the:

A. Superior laryngeal nerve

B. External laryngeal nerve

C. Recurrent laryngeal nerve

D. Phrenic nerve

The correct answer is C. Recurrent laryngeal nerve.

The motor innervation of the laryngeal muscles is a classic, high-yield rule in head and neck anatomy:

1. Recurrent Laryngeal Nerve (RLN) (Option C):

* Innervates ALL intrinsic laryngeal muscles (including the posterior cricoarytenoid, lateral cricoarytenoid, transverse/oblique arytenoids, and thyroarytenoid) EXCEPT the cricothyroid muscle.

* Also provides sensory innervation to the laryngeal mucosa below the vocal folds.

2. External Branch of the Superior Laryngeal Nerve (Option B):

* Innervates ONLY ONE intrinsic muscle: the cricothyroid muscle (which tenses the vocal cords to elevate voice pitch).

8. The inferior border of the parietal pleura (pleural reflection) is typically found at which rib level in the Mid-Axillary Line (MAL)?

A. Rib 6

B. Rib 8

C. Rib 10

D. Rib 12

C. Rib 10

• Explanation: The "Rule of 2s" for parietal pleural reflections:

• Midclavicular line: Rib 8

• Mid-axillary line: Rib 10

• Scapular/Posterior line: Rib 12

9. Which laryngeal muscle stretches or elongates the vocal cord, thereby raising the pitch of the voice?

A. Posterior cricoarytenoid

B. Lateral cricoarytenoid

C. Thyroarytenoid

D. Cricothyroid

D. Cricothyroid

• Explanation: The cricothyroid muscle tilts the cricoid cartilage upward and the thyroid cartilage forward, which tenses and elongates the vocal cords, producing high-pitched sounds.

10. The end of the trachea (bifurcation/carina) is typically located at the level of the:

A. Upper border of T4

B. Medial border of T4

C. Lateral border of T4

D. Inferior border of T4

D. Inferior border of T4

• Explanation: The trachea ends by bifurcating at the sternal angle of Louis, which corresponds posteriorly to the T4/T5 intervertebral disc level (or the inferior border of the T4 vertebra).

11. The muscle group that is primarily used to elevate the larynx during swallowing is the:

A. Infrahyoid muscles

B. Intrinsic laryngeal muscles

C. Suprahyoid muscles

D. Sternocleidomastoid

C. Suprahyoid muscles

• *Explanation: The suprahyoid muscles (mylohyoid, geniohyoid, stylohyoid, digastric) pull the hyoid bone and larynx upward (elevate) during swallowing to help seal the laryngeal inlet. (Infrahyoid muscles depress the larynx).*

12. The intercostal nerve runs primarily:

- A. Superior to the intercostal artery and vein (V-A-N order)*
- B. Between the external and internal intercostal muscles*
- C. Between the internal and innermost intercostal muscles*
- D. Posterior to the parietal pleura*

C. Between the internal and innermost intercostal muscles

** Explanation: The intercostal neurovascular bundle (VAN: Vein, Artery, Nerve from superior to inferior) travels in the costal groove specifically in the plane between the internal intercostal and innermost intercostal muscles**.*

13. Which major structure is found immediately posterior and superior to the root (hilum) of the right lung?

- A. Thoracic aorta*
- B. Azygos vein*
- C. Right phrenic nerve*

D. Esophagus

B. Azygos vein

• Explanation: The azygos vein ascends in the posterior mediastinum (posterior to the right root) and arches superiorly over the root of the right lung to empty into the Superior Vena Cava.

14. From superior to inferior, the typical arrangement of the main structures passing through the root (hilum) of the right lung is:

A. Pulmonary Artery, Bronchus, Pulmonary Vein

B. Pulmonary Vein, Pulmonary Artery, Bronchus

C. Eparterial Bronchus, Pulmonary Artery, Pulmonary Vein

D. Pulmonary Artery, Pulmonary Vein, Bronchus

C. Eparterial Bronchus, Pulmonary Artery, Pulmonary Vein

• Explanation: On the RIGHT side, because the bronchus branches early above the pulmonary artery, the arrangement from superior to inferior is:

1. Eparterial Bronchus (Most Superior)

2. Pulmonary Artery

3. Hyparterial Bronchus

4. Pulmonary Vein (Most Inferior)